

Reconfigurable Intelligent Surface-Aided Secret Key Generation in Multi-Cell Systems

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Abstract—Physical-layer key generation (PKG) exploits the reciprocity and randomness of wireless channels to generate a symmetric key between two legitimate communication ends. However, in multi-cell systems, PKG suffers from severe pilot contamination due to the reuse of pilots in different cells. In this paper, we invoke multiple reconfigurable intelligent surfaces (RISs) for adaptively shaping the environment and enhancing the PKG performance. To this end, we formulate an optimization problem to maximize the weighted sum key rate (WSKR) by jointly optimizing the precoding matrices at the base stations (BSs) and the phase shifts at the RISs. To address the non-convexity of the problem, we adopt an alternating optimization (AO)-based algorithm that divides the joint optimization problem into two subproblems. For the subproblem of precoding matrices, we apply the Lagrangian dual approach based on the Karush-Kuhn-Tucker (KKT) conditions. As for the subproblem of phase shifts, we adopt a projected gradient ascent (PGA) algorithm. Simulation results validate the effectiveness of the proposed scheme, demonstrating significant gains in WSKR. Moreover, compared with a single-RIS case, deploying

multiple RISs offer spatial diversity so as to improve the PKG performance of multicell systems.

Index Terms—Physical layer security, secret key generation, reconfigurable intelligent surface (RIS), multi-cell pilot contamination.

I. INTRODUCTION

THE ever-increasing connectivity among a large number of devices and the ubiquitous wireless communications have aroused great awareness to establish secure communication on the fly [1]. Conventionally, secure communication is guaranteed by applying cryptographic encryption mechanisms in the application layer [2]. Particularly, symmetric keys should be distributed to legitimate parties in these mechanisms before the actual communication takes place. However, existing cryptographic techniques face difficulties to realize secret key sharing in ad-hoc and mobile networks [3]. As an alternative, physical-layer key generation (PKG) exploits the intrinsic reciprocity and randomness of wireless channels to generate a pair of secret keys between the desired legitimate ends [4]. Furthermore, due to the existence of spatial decorrelation, an eavesdropper, Eve, cannot obtain any information about the generated keys if she locates more than half a wavelength away from the legitimate ends, i.e., Alice and Bob [5], [6], [7].

The process of PKG generally consists of four steps: channel sounding, quantization, information reconciliation, and privacy amplification [7]. During the channel probing step, Alice and Bob exchange pilots to acquire highly correlated channel estimations. Then, the extracted channel estimations are respectively quantized into bit sequences at Alice and Bob in the quantization step. Also, during the information reconciliation step, error-correcting codes are adopted to correct the mismatched bits between Alice and Bob. Finally, privacy amplification is employed to erase the bits that might have leaked information to Eve in the previous probing and reconciliation steps. From the above steps, it can be seen that PKG highly relies on the inherent randomness and reciprocity of wireless channels. However, the desired secret key rate may not be guaranteed in some harsh propagation environments, such as wave-blockage environments [8], [9]. Fortunately, reconfigurable intelligent surface (RIS), which has emerged as a disruptive wireless communication technology, holds tremendous potential to address this problem [10]. In fact, RIS is a planar surface comprising a large number of low-cost passive reflecting elements [10], [11], [12], [13]. These elements can independently adapt the phase shifts to collaboratively

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customize the wireless propagation environment such that the transmit rate can be enhanced [14]. Therefore, it is expected that deploying RIS in secure communication can facilitate the required PKG. In particular, when the direct link between Alice and Bob is blocked, RIS could shape a RIS-induced fluctuating channel to serve as a controllable randomizer for generating a secret key.

However, to fully unleash the potential of the RIS for improving PKG performance, the optimization of the phase shifts of RIS is required. To date, several studies have focused on the design of phase shifts of RIS [9], [15], [16], [17], [18], [19]. For instance, [9], [15], [16] studied the reflection coefficients optimization in single-input single-output (SISO) systems with only one legitimate user. In particular, the authors in [15] assumed the RIS-induced channel of the eavesdropper is independent from that of the legitimate ends. Based on this, they derived the expression of the key generation rate (KGR) capacity and optimized the on/off states of the RIS units. Furthermore, in [16], the authors considered RIS-assisted PKG with multiple non-colluding eavesdroppers. They designed a semidefinite relaxation (SDR) and successive convex approximation (SCA)-based algorithm to maximize the secret key capacity lower bound. Then, a RIS-assisted multiuser key generation scheme was studied in [9], where the RIS configuration was optimized to maximize the sum secret key rate of multiple users. On the other hand, [17], [18], [19] investigated the beamforming optimization in multiple-input single-output (MISO) systems. In particular, [17] proposed a low-complexity block successive upper-bound minimization (BSUM) with the mirror-prox method to optimize the reflective beamforming at the RIS and the transmit beamforming at the base station (BS), with the consideration of the spatial correlation at both the BS and the RIS. In addition, [18] treated the coupled precoding matrix and phase-shift matrix as an equivalent variable and designed a water-filling algorithm to acquire its optimal solution. Then, they recovered the two matrices from the optimized variable. Furthermore, to obtain a computationally efficient suboptimal solution to the non-convex problem in [18], the authors in [19] adopted a machine learning-based algorithm to achieve a higher KGR.

Nevertheless, all of these works i.e., [9], [15], [16], [17], [18], and [19], focus on the design of single-cell systems, while practical RIS-based PKG methods in multi-cell systems are still lacked. Indeed, in multi-cell systems, the same pilot pool is reused in different cells due to the limited time and frequency resources giving rise to the pilot contamination problem [20], [21], [22]. Over the last couple of years, considerable research efforts have been devoted for studying the impact of pilot contamination on spectral and energy efficiencies and the corresponding methods for alleviating the negative impacts caused by pilot interference [21], [22], [23], [24], [25]. However, all of these works aim for reducing the uplink channel estimation error that do not align with the goal of PKG which aims for improving the reciprocity between channel estimations in the uplink and downlink. Indeed, pilot contamination exists in both the uplink and downlink channel probing phase in PKG systems that introduces non-reciprocal interference to the channel estimations at the BSs and the user

terminals (UTs). Therefore, this channel asymmetry caused by multi-cell pilot contamination is expected to jeopardize the key generation performance. To tackle this problem, we propose to employ multiple RISs to assist the PKG in multi-cell networks. Specifically, by carefully altering the phase shifts introduced by multiple RISs, the inter-cell pilot interference reflected by the RISs and the interference in the direct channel can be harnessed such that they can be destructively superimposed at the desired communication nodes to minimize the interference power. Therefore, the proposed paradigm provides a new degrees of freedom (DoF) to facilitate multi-cell PKG in conjunction with the precoding matrices at the BSs. However, the BSs' precoding matrices and the RISs' phase shifts have to be jointly optimized to fully unleash the potential of RISs for effective KGR provisioning. More importantly, existing techniques in [9], [15], [16], [17], [18], and [19] cannot be directly applied to multi-cell PKG systems since the inter-cell pilot contamination is not taken into consideration.

To address the above issues, this paper investigates the PKG method in multi-cell systems. We introduce RISs to combat multi-cell pilot contamination by jointly optimizing the precoding matrices at the BSs and the phase shifts at the RISs. More specifically, the main contributions of this paper are as follows.

- We propose a novel RIS-aided multi-cell PKG framework based on the precoding matrices at the BSs and the phase shifts at the RISs. We then derive a closed-form KGR expression that facilitates the formulation of an optimization problem to maximize the weighted sum key rate (WSKR) of all the cells. We also prove that the KGR without RIS can be extremely low due to the inter-cell interference.
- To tackle the non-convex problem, we employ an alternating optimization (AO)-based algorithm to alternately obtain a high-quality solution. Specifically, a Lagrangian dual algorithm based on the KKT conditions is applied to design the precoding matrices at the BSs and a PGA algorithm is adopted to optimize the phase shifts at the RISs.
- Simulation results verify that the WSKR can be significantly enhanced by the proposed algorithm. Also, the performance gain increases with the number of RIS elements. Finally, distributed RISs, with each RIS being deployed at the cell boundaries, offer rich spatial diversity to improve the KGR performance of multicell systems.

Notations: In this paper, $\mathbb{C}^{A \times B}$ denotes the space of complex matrices of size $A \times B$. Matrices and vectors are denoted by boldface capital and lower-case letters, respectively. The imaginary unit of a complex number is denoted by $j = \sqrt{-1}$. $(\cdot)^T$ and $(\cdot)^H$ denote transpose and conjugate transpose, respectively. $\text{diag}(\mathbf{x})$ denotes a diagonal matrix whose diagonal elements are extracted from vector $\mathbf{x}$. $\text{vec}(\mathbf{X})$ denotes the vectorization of matrix $\mathbf{X}$. $\text{tr}(\cdot)$ represents the trace of a matrix. $\triangleq$ means "defined as". $\otimes$ and $\odot$ denote the Kronecker product and Khatri-Rao product, respectively. $\mathcal{I}(X; Y)$ and $\mathcal{H}(X, Y)$ are the mutual information and joint entropy of random variables X and Y , respectively. $\det(\cdot)$

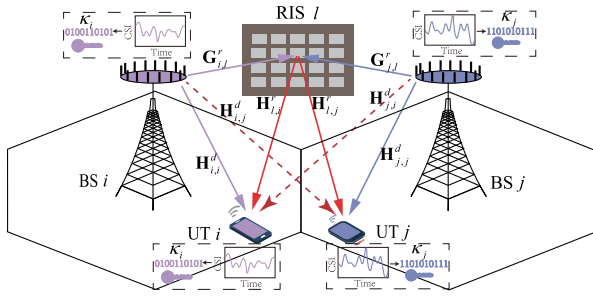


Fig. 1. The model of RIS-aided PKG in multi-cell systems.

is the matrix determinant. $[x]^+$ means $\max\{x, 0\}$. $\|\mathbf{X}\|_F$ is the Frobenius norm of matrix $\mathbf{X}$. $\mathbb{E}\{\cdot\}$ represents statistical expectation. $\lambda_\ell(\mathbf{X})$ is the ℓ -th largest eigenvalue of matrix $\mathbf{X}$. $\mathcal{O}(\cdot)$ is the big-O notation. $[\mathbf{X}]_{m:n,i}$ means the matrix consisting of rows m to n and column i of matrix $\mathbf{X}$. $\nabla f(\cdot)$ and $\partial f/\partial x$ are the gradient operator of function f . $\mathbf{I}_M$ denotes the identity matrix of dimension M .

II. RIS-BASED MULTI-CELL KEY GENERATION MODEL

As shown in Fig. 1, we consider a multi-cell PKG model constituted by K cells, each of which has an M_B antennas BS and an M_U antennas UT.¹ Under the time-division duplexing (TDD) protocol, the BS and UT in the k -th cell, i.e., BS k and UT k , $k \in \{1, \dots, K\}$, aim to generate a symmetric key κ_k by exploiting the reciprocity of the involved wireless channels. Also, we assume that different cells reuse the same frequency and pilot [20], [21], [22]. In this case, the channel estimations at the BSs and UTs are contaminated by the BSs and UTs from other cells, which is likely to degrade the PKG performance. To facilitate PKG, the system deploys L RISs with each RIS consisting of N passive reflection elements. Besides, there is a smart controller for coordinating the BSs and adapting the phase shifts of the RISs to enable effective secret key generation [27].

A. RIS-Based Channel Model

The direct channel from BS i to UT j is denoted as $\mathbf{H}_{i,j}^d \in \mathbb{C}^{M_B \times M_U}$, $i, j \in \{1, \dots, K\}$. When the RISs are introduced to the PKG system, they establish some indirect additional communication channels. Specifically, the channels from BS i to RIS l , $l \in \{1, \dots, L\}$, and from RIS l to UT j are denoted as $\mathbf{G}_{i,l}^r \in \mathbb{C}^{M_B \times N}$ and $\mathbf{H}_{l,j}^r \in \mathbb{C}^{N \times M_U}$, respectively. The diagonal phase-shifting matrix of RIS l is denoted by $\Phi_l = \text{diag}\{\mathbf{v}_l\}$, where $\mathbf{v}_l \in \mathbb{C}^{N \times 1}$ is the phase-shifting vector adopted at RIS l . Then, the equivalent downlink channel from BS i to UT j is expressed as²

$$\mathbf{H}_{i,j} = \mathbf{H}_{i,j}^d + \sum_{l=1}^L \mathbf{G}_{i,l}^r \Phi_l \mathbf{H}_{l,j}^r = \mathbf{H}_{i,j}^d + \mathbf{G}_i^r \Phi \mathbf{H}_j^r, \quad (1)$$

¹The pilot sequences assigned to different UTs in each cell are assumed to be orthogonal to each other to avoid potential intra-cell interference such that pilot contamination only exists among inter-cell UTs [26].

²Note that different multipath delays incurred by the pilot delays are ignored because the pilots are assumed to be synchronized to maximize the power of multi-cell interference that serves as a worst case scenario [21].

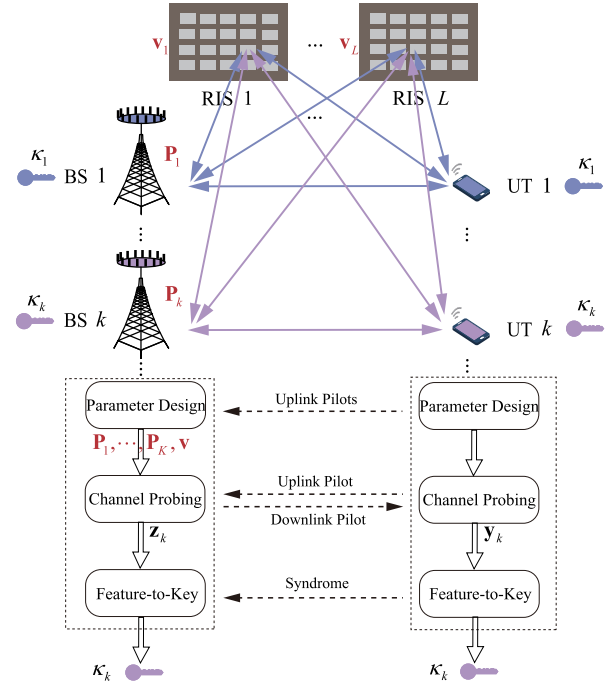


Fig. 2. The proposed RIS-aided PKG framework in multi-cell systems.

where $\mathbf{G}_i^r = [\mathbf{G}_{i,1}^r, \dots, \mathbf{G}_{i,L}^r]$, $\mathbf{H}_j^r = [(\mathbf{H}_{1,j}^r)^H, \dots, (\mathbf{H}_{L,j}^r)^H]^H$, and $\Phi = \text{diag}(\Phi_1, \dots, \Phi_L)$.

From (1), the channel between BS i and UT j depends on the wireless channels and phase-shifting matrix Φ . This provides a new DoF for optimizing the key generation performance. Also, the multi-antenna BSs have the intrinsic capability of signal processing in the spatial domain. Hence, we next establish a new PKG framework to exploit the spatial diversity of the multi-antenna BSs and the environment-controlling characteristic of the RISs to enable the multi-cell secret key generation.

B. RIS-Aided PKG Framework

As shown in Fig. 2, the proposed framework consists of three phases. Firstly, during the parameter design phase, the BSs design the precoding matrices and phase-shifting vector. Then, they convey the phase-shifting vector to the RISs controller to configure the RISs reflection matrix. Secondly, the BSs and UTs acquire the channel estimations by channel probing. Finally, feature-to-key processing is performed to convert the channel estimations into secret keys [6], [9]. The specific procedures are shown as follows.

- 1) *Parameter Design*: In this step, the BSs design the precoding matrices $\mathbf{P}_k$, $k \in \{1, \dots, K\}$, and phase shifts Φ according to some statistical CSI. In particular, the UTs in all the cells transmit orthogonal sounding signals such that the BSs can obtain the channel covariance matrices between the BSs and the UTs. Then, the BSs design the parameters $\mathbf{P}_k$ and $\mathbf{v}$ via the proposed optimization algorithm, which will be elaborated in Section IV.
- 2) *Channel Probing*: In this step, the BSs and UTs probe the channel alternately and extract the reciprocal channel characteristics with the help of the precoding

matrices and reflection coefficients. In the downlink, each BS transmits the downlink pilot signals processed by the optimized precoding matrix $\mathbf{P}_k$. Then, the UTs estimate the combined channels from the received signals. In the uplink, each UT transmits the same pilot signals. The pilot is reflected by the RISs and received by the BSs. Then, the BSs estimate the channels from the received signals and utilize the precoding matrices $\mathbf{P}_k$ to establish the combined channels.

- 3) *Feature-to-Key Processing*: Once the BSs and UTs acquire the channel estimations, they employ some quantization method to generate raw key bits. Finally, information reconciliation and privacy amplification steps are performed to produce the secret keys.

Remark 1: In the parameter design step, the prior information to design the precoding matrices and phase shifts are the channel covariance matrices.³ Upon the parameter design phase is completed, the BSs and UTs can perform multiple channel probing rounds within some channel coherence times. Each channel probing round is completed within one coherence time to guarantee the similarity of the uplink and downlink channel estimations. Meanwhile, different channel probing rounds should be performed in different coherence times to introduce randomness to the secret keys.

In this paper, we first focus on the channel probing step, where the multi-cell pilot contamination exists and then propose an algorithm to design the precoding matrices and phase shifts to improve the PKG performance.

C. Signal Presentation of the Channel Probing

In the PKG system, the BSs and UTs first perform channel probing to acquire the reciprocal channel estimation. The process of channel probing is described as follows.

The channel sounding step consists of two phases, i.e., the downlink phase and the uplink phase, respectively. In the downlink, BS k transmits the downlink publicly known pilot $\mathbf{X} \in \mathbb{C}^{M_B \times M_B}$ with $\mathbf{X}\mathbf{X}^H = \mathbf{I}_{M_B}$. Then, the signals received at UT k is

$$(\mathbf{Y}_k^d)^T = \mathbf{H}_{k,k}^T \mathbf{P}_k^T \mathbf{X} + \underbrace{\sum_{i=1, i \neq k}^K \mathbf{H}_{i,k}^T \mathbf{P}_i^T \mathbf{X}}_{\text{Signals from BSs in other cells}} + (\mathbf{N}_k^d)^T, \quad (2)$$

where $\mathbf{P}_i \in \mathbb{C}^{M_B \times M_B}$ represents the precoding matrix at BS i [6], [30],⁴ $\mathbf{N}_k^d \in \mathbb{C}^{M_B \times M_B}$ is the complex Gaussian noise at UT k with zero mean and unit variance. After the standard

least-squares (LS) channel estimation⁵ [9], [16], UT k obtains

$$\mathbf{Y}_k = \mathbf{X}^* \mathbf{Y}_k^d = \mathbf{P}_k \mathbf{H}_{k,k} + \sum_{i=1, i \neq k}^K \mathbf{P}_i \mathbf{H}_{i,k} + \mathbf{X}^* \mathbf{N}_k^d. \quad (3)$$

In the uplink phase, the UTs simultaneously transmit the pilot $\mathbf{S}$ with $\mathbf{S}\mathbf{S}^H = \mathbf{I}_{M_U}$ and the signal received at BS k is expressed as

$$\mathbf{Z}_k^u = \mathbf{H}_{k,k} \mathbf{S} / \sqrt{M_U} + \underbrace{\sum_{j=1, j \neq k}^K \mathbf{H}_{k,j} \mathbf{S} / \sqrt{M_U}}_{\text{Signals from UTs in other cells}} + \mathbf{N}_k^u, \quad (4)$$

where $\mathbf{N}_k^u$ is the complex Gaussian noise zero mean and unit variance. Then, BS k performs the LS channel estimation as

$$\tilde{\mathbf{Z}}_k = \mathbf{Z}_k^u \mathbf{S}^H = \mathbf{H}_{k,k} / \sqrt{M_U} + \sum_{j=1, j \neq k}^K \mathbf{H}_{k,j} / \sqrt{M_U} + \mathbf{N}_k^u \mathbf{S}^H. \quad (5)$$

After that, BS k multiplies the channel estimation with the precoding matrix $\mathbf{P}_k$ that yields

$$\begin{aligned} \mathbf{Z}_k &= \mathbf{P}_k \tilde{\mathbf{Z}}_k \\ &= \mathbf{P}_k \mathbf{H}_{k,k} / \sqrt{M_U} + \mathbf{P}_k \sum_{j=1, j \neq k}^K \mathbf{H}_{k,j} / \sqrt{M_U} + \mathbf{P}_k \mathbf{N}_k^u \mathbf{S}^H. \end{aligned} \quad (6)$$

Finally, BS k and UT k vectorize the channel matrices $\mathbf{Z}_k$ and $\mathbf{Y}_k$ as

$$\begin{aligned} \mathbf{z}_k &= \text{vec}(\mathbf{Z}_k) \\ &= \bar{\mathbf{P}}_k \mathbf{h}_{k,k} / \sqrt{M_U} + \bar{\mathbf{P}}_k \sum_{j=1, j \neq k}^K \mathbf{h}_{k,j} / \sqrt{M_U} + \bar{\mathbf{P}}_k \mathbf{n}_k^u, \end{aligned} \quad (7)$$

$$\mathbf{y}_k = \text{vec}(\mathbf{Y}_k) = \bar{\mathbf{P}}_k \mathbf{h}_{k,k} + \sum_{i=1, i \neq k}^K \bar{\mathbf{P}}_i \mathbf{h}_{i,k} + \mathbf{n}_k^d, \quad (8)$$

respectively, where $\bar{\mathbf{P}}_i \triangleq \mathbf{I}_{M_U} \otimes \mathbf{P}_i$, $\mathbf{h}_{i,j} \triangleq \text{vec}(\mathbf{H}_{i,j})$, $\mathbf{n}_k^d \triangleq \text{vec}(\mathbf{X}^* \mathbf{N}_k^d)$, and $\mathbf{n}_k^u \triangleq \text{vec}(\mathbf{N}_k^u \mathbf{S}^H)$.

It can be observed from (7) and (8) that the extracted channel features at BS k and UT k include the reciprocal component $\bar{\mathbf{P}}_k \mathbf{h}_{k,k}$, the inter-cell interference, and as well as the noise component. These interference terms are not perfectly reciprocal. In particular, in the k -th cell, the interference in the uplink, i.e., $\bar{\mathbf{P}}_k \sum_{j=1, j \neq k}^K \mathbf{h}_{k,j}$, is the product of $\bar{\mathbf{P}}_k$ and the aggregated channels from UT j to BS k , while in the downlink, the interference, i.e., $\sum_{i=1, i \neq k}^K \bar{\mathbf{P}}_i \mathbf{h}_{i,k}$, is the superimposed channels from BS i to UT k . Fortunately, these components are influenced by the precoding matrices $\mathbf{P}_i, i \in \{1, \dots, K\}$, at the BSs and the RIS phase-shifting vector $\mathbf{v}$. Next, we will analyze the impact of $\mathbf{P}_i$ and $\mathbf{v}$ on KGR.

⁵The LS is adopted since it has been widely implemented in practical systems [31].

³Due to the fact that the covariance matrices alter slowly across time in dense scattering environments [28], we can obtain these matrices from the previous several time slots. As an example, the covariance matrices can be obtained by performing T_e rounds of uplink channel probing via the method in [29] and the required pilot overhead is calculated as $T_e K M_U (N L + 1)$.

⁴It is noted that the proposed design can also be extended to the case where $\mathbf{P}_i \in \mathbb{C}^{M_F \times M_B}$ with $M_F < M_B$. In this case, the expression (12) provides an upper bound for the KGR, which can be optimized via the proposed algorithm in this paper.

III. SECRET KEY RATE ANALYSIS

In this section, based on the channel estimations shown in Section II, we derive the closed-form KGR expression and formulate an optimization problem with respect to (w.r.t.) the variables $\mathbf{P}_i, i \in \{1, \dots, K\}$, and $\mathbf{v}$.

A. KGR Analysis

To evaluate the key generation performance, we derive the KGR expression in the following. The KGR is a comprehensive metric to evaluate the consistency, rate, and randomness of the secret key [32]. The KGR between UT k and BS k is defined as the mutual information between their channel estimations, which is expressed as

$$R_k = \mathcal{I}(\mathbf{y}_k; \mathbf{z}_k). \quad (9)$$

Given the combined channel gains $\mathbf{y}_k$ and $\mathbf{z}_k$ in (7) and (8), the closed-form expression of the KGR can be derived as follows.

Theorem 1: The KGR of UT k and BS k is given by

$$R_k = \ln \det \left(\sum_{i=1}^K \bar{\mathbf{P}}_i \mathbf{M}_{i,k} \bar{\mathbf{P}}_i^H + \mathbf{I}_M \right) - \ln \det \left(\sum_{i=1}^K \bar{\mathbf{P}}_i \mathbf{M}_{i,k} \bar{\mathbf{P}}_i^H + \mathbf{I}_M - \bar{\mathbf{P}}_k \mathbf{N}_{k,k} \bar{\mathbf{P}}_k^H \right), \quad (10)$$

where

$$\begin{aligned} \mathbf{M}_{i,k} &= \mathbf{R}_{i,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \mathbf{R}_{i,k}^r (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M)^H, \\ \mathbf{N}_{k,k} &= (\mathbf{R}_{k,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \mathbf{R}_{k,k}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M)) \\ &\quad \times \left[\sum_{j=1}^K \mathbf{R}_{k,j}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \sum_{j=1}^K \mathbf{R}_{k,j}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \right. \\ &\quad \left. + \mathbf{I}_M M_U \right]^{-1} (\mathbf{R}_{k,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \mathbf{R}_{k,k}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M)). \end{aligned} \quad (12)$$

Also, $M \triangleq M_B M_U$, $\mathbf{R}_{i,j}^d \triangleq \mathbb{E} \left\{ \text{vec}(\mathbf{H}_{i,j}^d) \text{vec}(\mathbf{H}_{i,j}^d)^H \right\}$ and $\mathbf{R}_{i,j}^r \triangleq \mathbb{E} \left\{ \text{vec}(\mathbf{H}_{i,j}^r) \text{vec}(\mathbf{H}_{i,j}^r)^H \right\}$ are the channel covariance of the direct channel and the RISs-involved channel between BS i and UT j , respectively, where $\mathbf{H}_{i,j}^r \triangleq (\mathbf{H}_j^r)^T \odot \mathbf{G}_i^r$. The variable is defined as $\bar{\mathbf{v}} \triangleq \mathbf{v}^*$ for the sake of notational simplicity.

Proof: Please see Appendix A. $\square$

Remark 2: When the precoding matrices at the BSs are not optimized, i.e., $\mathbf{P}_i = \sqrt{\frac{P_i}{M_B}} \mathbf{I}_{M_B}$, and the RISs are not deployed, the KGR between BS k and UT k is given by

$$\begin{aligned} R_k &= \ln \det \left(\sum_{i=1}^K \frac{P_i}{M_B} \mathbf{R}_{i,k}^d + \mathbf{I}_M \right) \\ &\quad - \ln \det \left(\sum_{i=1}^K \frac{P_i}{M_B} \mathbf{R}_{i,k}^d + \mathbf{I}_M - \frac{P_k}{M_B} \mathbf{R}_{k,k}^d \right. \\ &\quad \left. \times \left(\sum_{j=1}^K \frac{P_j}{M_B} \mathbf{R}_{k,j}^d + M_U \mathbf{I}_M \right)^{-1} \mathbf{R}_{k,k}^d \right), \end{aligned} \quad (13)$$

where P_i denotes the transmit power at BS i . Furthermore, we present a proposition of the KGR performance as follows.

Proposition 1: R_k decreases monotonically with increasing $P_i, i \neq k$, and the eigenvalues of $\mathbf{R}_{i,k}^d$ and $\mathbf{R}_{k,j}^d, j \neq k$. Also, R_k approaches 0 when these values are sufficiently large.

Proof: Please see Appendix B. $\square$

Proposition 1 proves that the KGR between BS k and UT k decreases with the transmit power of BSs in other cells, i.e., $P_i, i \neq k$, and the channel gain of the interference channels, i.e., $\lambda_\ell(\mathbf{R}_{i,k}^d), i \neq k$, and $\lambda_\ell(\mathbf{R}_{k,j}^d), j \neq k$, $\ell \in \{1, \dots, M\}$. In addition, R_k tends to 0 when these interference is sufficiently large. Therefore, the crux of the multi-cell PKG is to effectively mitigate the pilot contamination in the uplink and downlink.

From (10), the KGR R_k is determined by the statistical channel information (CSI), i.e., the channel covariance matrices, and the optimization variables, i.e., the precoding matrices and the phase-shifting vector. We assume that the covariance matrices are perfectly known by the BSs,⁶ and we focus on the design of precoding matrices and phase-shifting vector.

Now, we jointly design the precoding matrices and phase shifts to improve the key generation performance. Specifically, we aim for maximizing the WSKR of all the cells by jointly optimizing the $\mathbf{P}_i, i \in \{1, \dots, K\}$, and $\bar{\mathbf{v}}$, while guaranteeing the total power constraint of each BS and the unit modulus of the reflection coefficient of each RIS element. Specifically, the optimization problem can be formulated as

$$\begin{aligned} &\underset{\mathbf{P}_i, i \in \{1, \dots, K\}, \bar{\mathbf{v}}}{\text{maximize}} \quad \sum_{k=1}^K w_k R_k \\ &\text{s.t. C1: } \|\mathbf{P}_i\|_F^2 \leq P_A, \forall i \in \{1, \dots, K\}, \\ &\quad \text{C2: } |v_n| = 1, \forall n \in \{1, \dots, NL\}, \end{aligned} \quad (14)$$

where $w_k \geq 0$ denotes the weight for the k -th cell that represents the priority of the corresponding BS and UT. Constraint C1 represents the power of the precoding matrices at each BS should be less than the maximum transmit power P_A [30]. Constraint C2 represents the modulus constraint of each reflection coefficient at the RISs [33].

Note that the objective function WSKR is non-convex w.r.t. $\mathbf{P}_i$ and $\bar{\mathbf{v}}$ since these variables are coupled in the matrix inversion operation. Additionally, $\mathbf{P}_i$ is coupled with the identity matrix in $\bar{\mathbf{P}}_i$ and the unit-modulus constraint C2 is also non-convex. Hence, Problem (14) is generally NP-hard and difficult to find the globally optimal solution [34]. In the next section, we first denote $\bar{\mathbf{P}}_i$ as a new variable and apply an AO-based algorithm to alternately solve for $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$ and $\bar{\mathbf{v}}$ while fixing one of the variable sets. Then, we extract the optimal $\mathbf{P}_i$ from $\bar{\mathbf{P}}_i$.

IV. AO-BASED ALGORITHM FOR MAXIMUM WSKR

In this section, we first take $\bar{\mathbf{P}}_i = \mathbf{I}_{M_U} \otimes \mathbf{P}_i$ as the new variable and provide an AO-based algorithm to tackle Problem (14). Specifically, to tackle the coupling of the $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$ and $\bar{\mathbf{v}}$, we divide the joint problem into

⁶The impact of imperfect statistical CSI on PKG performance is left to future work.

two sub-problems, each of which optimizes $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$ or $\bar{\mathbf{v}}$. For the sub-problem for $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$, we apply a Lagrange multiplier method based on KKT conditions, while the sub-problem for $\bar{\mathbf{v}}$ is solved by the PGA algorithm. Finally, the optimal $\mathbf{P}_i$ can be extracted from each $\bar{\mathbf{P}}_i$ by solving a convex problem.

A. Optimization of Precoding Matrices $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$

We first present the optimization of precoding matrices $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$ for given $\bar{\mathbf{v}}$. By denoting $\bar{\mathbf{P}} = [\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K]$, the sub-problem of $\bar{\mathbf{P}}$ optimization is given by

$$\begin{aligned} \underset{\bar{\mathbf{P}}}{\text{maximize}} \quad & \bar{g}(\bar{\mathbf{P}}) = \sum_{k=1}^K w_k (\ln \det(\bar{\mathbf{P}} \mathbf{M}_k \bar{\mathbf{P}}^H + \mathbf{I}_M) \\ & - \ln \det(\bar{\mathbf{P}} \mathbf{N}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)) \\ \text{s.t.} \quad & \text{C3: } \|\bar{\mathbf{P}}_i\|_F^2 \leq M_U P_A, \forall i \in \{1, \dots, K\}, \end{aligned} \quad (15)$$

where $\mathbf{M}_k = \text{diag}(\mathbf{M}_{1,k}, \dots, \mathbf{M}_{K,k})$ and $\mathbf{N}_k = \text{diag}(\mathbf{M}_{1,k}, \dots, \mathbf{M}_{k,k} - \mathbf{N}_{k,k}, \dots, \mathbf{M}_{K,k})$.

It can be observed that Problem (15) is still non-convex since the objective function is in high-order w.r.t. $\bar{\mathbf{P}}$ and in the form of $\ln \det(\cdot) - \ln \det(\cdot)$. Hence, it is difficult to acquire the optimal solution by applying the min-max representation [35] or obtain a suboptimal via the minorize-maximization (MM) algorithm [36]. To address this problem, we employ the Lagrange multiplier method to obtain a KKT solution. Specifically, by introducing the Lagrange multipliers $\lambda_i, i = 1, \dots, K$, the Lagrangian function of Problem (15) is

$$\begin{aligned} \mathcal{L}(\bar{\mathbf{P}}_i, \lambda_i) = & \sum_{k=1}^K w_k (\ln \det(\bar{\mathbf{P}} \mathbf{M}_k \bar{\mathbf{P}}^H + \mathbf{I}_M) \\ & - \ln \det(\bar{\mathbf{P}} \mathbf{N}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)) - \text{tr}[\bar{\mathbf{P}} \mathbf{A} \bar{\mathbf{P}}^H] \\ & + M_U P_A \sum_{i=1}^K \lambda_i, \end{aligned} \quad (16)$$

where $\mathbf{A} = \text{diag}(\lambda_1 \mathbf{I}_M, \dots, \lambda_K \mathbf{I}_M)$ and $\lambda_i \geq 0, i \in \{1, \dots, K\}$, is associated with the power constraint in C3 at BS i . By analyzing the KKT condition, we can characterize the structure of the suboptimal solution to $\bar{\mathbf{P}}$ in the following theorem.

Theorem 2: The KKT condition for Problem (15) is given by

$$\begin{aligned} & \text{vec}(\bar{\mathbf{P}}) \\ & = \left(\mathbf{A}^T \otimes \mathbf{I}_M + \sum_{k=1}^K w_k \left(\mathbf{N}_k^T \otimes (\mathbf{I}_M + \bar{\mathbf{P}} \mathbf{N}_k \bar{\mathbf{P}}^H)^{-1} \right) \right)^{-1} \\ & \quad \times \sum_{k=1}^K w_k \left(\left(\mathbf{M}_k^T \otimes (\bar{\mathbf{P}} \mathbf{M}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)^{-1} \right) \text{vec}(\bar{\mathbf{P}}) \right). \end{aligned} \quad (17)$$

Proof: Please see Appendix C. $\square$

Next, the value of $\lambda_i, i \in \{1, \dots, K\}$, should be chosen for ensuring the power constraint at each BS in (14) is satisfied, which means

$$\lambda_i (\text{tr}(\bar{\mathbf{P}}_i \bar{\mathbf{P}}_i^H) - M_U P_A) = 0. \quad (18)$$

To this end, recalling the KKT condition in (17), we can rewrite the $\text{vec}(\bar{\mathbf{P}})$ as

$$\begin{aligned} & \text{vec}(\bar{\mathbf{P}}) \\ & = (\text{diag}(\lambda_1 \mathbf{I}_{M^2}, \dots, \lambda_K \mathbf{I}_{M^2}) + \mathbf{F})^{-1} \mathbf{a} \\ & = \text{diag}((\lambda_1 \mathbf{I}_{M^2} + \mathbf{F}_1)^{-1}, \dots, (\lambda_K \mathbf{I}_{M^2} + \mathbf{F}_K)^{-1}) \mathbf{a}, \end{aligned} \quad (19)$$

where

$$\begin{aligned} \mathbf{F} & = \sum_{k=1}^K w_k \left(\mathbf{N}_k^T \otimes (\mathbf{I}_M + \bar{\mathbf{P}} \mathbf{N}_k \bar{\mathbf{P}}^H)^{-1} \right) \\ & = \text{diag}(\mathbf{F}_1, \dots, \mathbf{F}_K), \end{aligned} \quad (21)$$

and $\mathbf{a} = \sum_{k=1}^K w_k \left(\left(\mathbf{M}_k^T \otimes (\bar{\mathbf{P}} \mathbf{M}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)^{-1} \right) \text{vec}(\bar{\mathbf{P}}) \right) = [\mathbf{a}_1^T, \dots, \mathbf{a}_K^T]^T$. Thus, we can express the $\bar{\mathbf{P}}_i$ at BS i as

$$\text{vec}(\bar{\mathbf{P}}_i) = (\lambda_i \mathbf{I}_{M^2} + \mathbf{F}_i)^{-1} \mathbf{a}_i. \quad (22)$$

By performing eigenvalue decomposition on matrix $\mathbf{F}_i$ as $\mathbf{F}_i = \mathbf{D}_i \text{diag}(\mathbf{f}_i) \mathbf{D}_i^H$, we have

$$\phi_i(\lambda_i) = \text{vec}(\bar{\mathbf{P}}_i)^H \text{vec}(\bar{\mathbf{P}}_i) \quad (23)$$

$$\begin{aligned} & = \mathbf{a}_i^H \mathbf{D}_i (\lambda_i \mathbf{I}_{M^2} + \text{diag}(\mathbf{f}_i))^{-1} (\lambda_i \mathbf{I}_{M^2} + \text{diag}(\mathbf{f}_i))^{-1} \\ & \quad \times \mathbf{D}_i^H \mathbf{a}_i = \sum_{m=1}^{M^2} \frac{b_{i,m}^2}{(\lambda_i + f_{i,m})^2}, \end{aligned} \quad (24)$$

where $\mathbf{f}_i = [f_{i,1}, \dots, f_{i,M^2}]^T$ and $\mathbf{D}_i^H \mathbf{a}_i = [b_{i,1}, \dots, b_{i,M^2}]^T$. Since $f_{i,m} \geq 0$, it can be verified that $f_i(\lambda_i)$ is a monotonically decreasing function w.r.t. λ_i .

Therefore, if $\phi_i(0) \leq M_U P_A$, the optimized $\bar{\mathbf{P}}_i$ is $\text{vec}(\bar{\mathbf{P}}_i^{\text{KKT}}) = \mathbf{F}_i^{-1} \mathbf{a}_i$ when λ_i is set to 0. On the other hand, if $\phi_i(0) > M_U P_A$, there exists a $\bar{\lambda}_i$ that satisfies $\phi_i(\bar{\lambda}_i) = M_U P_A$. To find $\bar{\lambda}_i$, we can apply the bisection-based search method and the upper bound of λ_i is set as

$$\lambda_i^{\text{ub}} = \sqrt{\frac{\sum_{m=1}^M b_{i,m}^2}{M_U P_A}}, \quad (25)$$

since

$$\phi_i(\lambda_i) < \sum_{m=1}^M \frac{b_{i,m}^2}{(\lambda_i^{\text{ub}})^2} = M_U P_A. \quad (26)$$

In addition, since the primal problem in (15) is non-convex, the solution obtained by the Lagrange multiplier method is determined by the initial value of $\bar{\mathbf{P}}_k$, i.e., $\bar{\mathbf{P}}_k^{(0)}$. Therefore, according to [37], we set $\bar{\mathbf{P}}_k^{(0)}$ to the optimal precoding matrix for the single-cell scenario. Specifically, by performing eigenvalue decomposition, we have $\bar{\mathbf{P}}_k^{(0)} = \mathbf{U}_{\mathbf{P}_k} \text{diag}(\mathbf{p}_k) \mathbf{U}_{\mathbf{P}_k}^H$, where $\mathbf{U}_{\mathbf{P}_k} \in \mathbb{C}^{M \times M}$ and $\mathbf{p}_k = [p_1, \dots, p_M]^T$ denote the matrix containing the eigenvectors and the vector containing the corresponding eigenvalue sorted in descending orders, respectively. Then, by denoting $\mathbf{R}_{k,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \mathbf{R}_{k,k}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M) = \mathbf{U}_{\mathbf{Q}_k} \text{diag}(\mathbf{q}_k) \mathbf{U}_{\mathbf{Q}_k}^H$, where $\mathbf{U}_{\mathbf{Q}_k}$ and $\mathbf{q}_k = [q_1, \dots, q_M]^T$ are the matrix containing the eigenvectors and the vector containing the corresponding

eigenvalue sorted in descending orders, respectively, the optimal $\mathbf{U}_{\mathbf{P}_k}$ is equal to $\mathbf{U}_{\mathbf{Q}_k}$ and the optimal $\mathbf{p}_k$ is given by the waterfilling-based power allocation

$$p_m(\mu) = \left[-\frac{2q_m + 1}{2q_m(q_m + 1)} + \sqrt{\frac{1 + q_m}{4} + \frac{1}{\mu \ln 2}} \right]^+,$$

$$m \in \{1, \dots, M\}, \quad (27)$$

where the waterfilling level $\mu > 0$ is chosen such that

$$\sum_{m=1}^M p_m = M_{\text{U}} P_A \quad (28)$$

is satisfied.

The overall algorithm to optimize $\bar{\mathbf{P}}$ is summarized in Algorithm 1.

Algorithm 1 The Lagrangian Dual Algorithm for Problem (15)

Require: $w_k, \mathbf{M}_k, \mathbf{N}_k, k = 1, \dots, K, P_A, \epsilon$.

1: Set: $t = 0$ (iteration index).

2: Initial: $\bar{\mathbf{P}}^{(0)} = [\bar{\mathbf{P}}_1^{(0)}, \dots, \bar{\mathbf{P}}_K^{(0)}]$.

3: **repeat**

4: Calculate $\bar{\mathbf{P}}^{(t+1)}$ as

$$\text{vec}(\bar{\mathbf{P}}^{(t+1)}) = \left(\mathbf{\Lambda}^T \otimes \mathbf{I}_M + \sum_{k=1}^K w_k (\mathbf{N}_k^T \otimes (\mathbf{I}_M + \bar{\mathbf{P}}^{(t)} \mathbf{N}_k (\bar{\mathbf{P}}^{(t)})^H)^{-1}) \right)^{-1}$$

$$\times \sum_{k=1}^K w_k \left((\mathbf{M}_k^T \otimes (\bar{\mathbf{P}}^{(t)} \mathbf{M}_k (\bar{\mathbf{P}}^{(t)})^H + \mathbf{I}_M)^{-1}) \text{vec}(\bar{\mathbf{P}}^{(t)}) \right). \quad (29)$$

5: Set: $i = 1$ (BS index).

6: **repeat**

7: Initialize the bounds λ_i^{ub} and λ_i^{lb} .

8: **repeat**

9: If $\phi_i(0) \leq P_A$ holds, $\text{vec}(\bar{\mathbf{P}}_i^{\text{KKT}}) = \mathbf{F}_i^{-1} \mathbf{a}_i$;
otherwise, go to the next step.

10: Calculate $\lambda_i = (\lambda_i^{\text{ub}} + \lambda_i^{\text{lb}})/2$.

11: If $\phi_i(\lambda_i) \leq P_A$, set $\lambda_i^{\text{ub}} = \lambda_i$; Otherwise, set

12: $\lambda_i^{\text{lb}} = \lambda_i$.

13: **until** $|\lambda_i^{\text{ub}} - \lambda_i^{\text{lb}}| \leq \epsilon$.

14: **until** $i = K$.

15: Set $t = t + 1$ and calculate the secret key rate $\bar{g}(\bar{\mathbf{P}}^{(t+1)})$.

16: **until** $\frac{|\bar{g}(\bar{\mathbf{P}}^{(t+1)}) - \bar{g}(\bar{\mathbf{P}}^{(t)})|}{\bar{g}(\bar{\mathbf{P}}^{(t)})} \leq \epsilon$.

B. Optimization of Phase Shifts $\bar{\mathbf{v}}$

Next, we present the optimization of phase shift vector $\bar{\mathbf{v}}$ when $\{\bar{\mathbf{P}}_1, \dots, \bar{\mathbf{P}}_K\}$ are fixed. To facilitate the algorithm design, we can first formulate the sub-problem for the RISs as

$$\underset{\bar{\mathbf{v}}}{\text{maximize}} \quad \bar{g}(\bar{\mathbf{v}}) = \sum_{k=1}^K w_k R_k(\bar{\mathbf{v}})$$

$$\text{s.t.} \quad \text{C2}, \quad (30)$$

where

$$R_k(\bar{\mathbf{v}}) = \ln \det(\mathbf{I}_k) - \ln \det(\mathbf{I}_k - (\bar{\mathbf{P}}_k \mathbf{R}_{k,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \times (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_k) \mathbf{R}_{k,k}^r (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M)^H) \\ \times \left(\sum_{j=1}^K \mathbf{R}_{k,j}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \sum_{j=1}^K \mathbf{R}_{k,j}^r (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M)^H + \mathbf{I}_M M_{\text{U}} \right)^{-1} (\mathbf{R}_{k,k}^d \bar{\mathbf{P}}_k^H + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \times \mathbf{R}_{k,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_k)^H (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M)^H) \Big), \quad (31)$$

where $\mathbf{I}_k = \sum_{i=1}^K \bar{\mathbf{P}}_i \mathbf{R}_{i,k}^d \bar{\mathbf{P}}_i^H + \mathbf{I}_M + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \sum_{i=1}^K (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i) \mathbf{R}_{i,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i)^H (\bar{\mathbf{v}} \otimes \mathbf{I}_M)$. From (31), the objective function is intractable w.r.t. $\bar{\mathbf{v}}$. The unit modulus constraint C2 is also non-convex and there is no general approach to solve unit modulus constrained problems optimally. Therefore, we adopt the PGA algorithm to find a stationary solution of Problem (30) [38]. At each iteration, PGA projects the solution onto the closest feasible point satisfying the unit-modulus constraint. Specifically, at iteration t , we first calculate the conjugate gradient $\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}})$ as the ascent direction to guarantee the increase of the objective function. The closed-form expression of the gradient is derived as follows.

Theorem 3: The Euclidean gradient of function $g(\bar{\mathbf{v}})$ w.r.t. $\bar{\mathbf{v}}^*$ is given by

$$\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}}) = \sum_{k=1}^K w_k \left(\sum_{m=1}^M \bar{\mathbf{g}}_{k,m} \right), \quad (32)$$

where $\bar{\mathbf{g}}_{k,m} = [\mathbf{G}_k]_{(m-1)NL+1:mNL,m}$ and

$$\mathbf{G}_k = \mathbf{K}_{MNL} \left(\sum_{i=1}^K (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i) \mathbf{R}_{i,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i)^H \times (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \mathbf{R}_{\mathbf{y}_k}^{-1} - \sum_{i=1}^K (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i) \mathbf{R}_{i,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i)^H \times (\bar{\mathbf{v}} \otimes \mathbf{I}_M) (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k})^{-1} \right. \\ \left. + (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_k) \mathbf{R}_{k,k}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} \times (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k})^{-1} - \sum_{j=1}^K \mathbf{R}_{k,j}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \right. \\ \left. \times \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k})^{-1} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \right. \\ \left. + \mathbf{R}_{k,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_k)^H (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \times (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k})^{-1} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1} \right) \mathbf{K}_M, \quad (33)$$

where $\mathbf{K}_I \in \mathbb{C}^{I \times I}, I \in \{MNL, M\}$, is the commutation matrix [39].

Proof: Please see Appendix D. $\square$

Then, the next iteration point is calculated as $\bar{\mathbf{v}}^{(t+1)} = \exp(j \arg(\bar{\mathbf{v}}^{(t)} + \mu^{(t)} \nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}}^{(t)})))$, where $\mu^{(t)}$ is the step size computed by the backtracking line search [34] and the $\arg(\cdot)$ operation is adopted for satisfying the unit-modulus constraint.

The overall algorithm to optimize $\bar{\mathbf{v}}$ is summarized as Algorithm 2. The objective values of Problem (30) are non-decreasing since the search direction is set as the steepest ascent direction $\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}})$ and the backtracking line search is employed to find a suitable step size [34]. In addition, since the solution set for $\bar{\mathbf{v}}$ is compact, the maximum value of WSKR is bounded. Therefore, the objective value converges over iterations.

Algorithm 2 The PGA Algorithm for Problem (30)

Require: $\mathbf{R}_{i,j}^d, \mathbf{R}_{i,j}^r, i, j = 1, \dots, K, \epsilon$.

- 1: Set: $t = 0$ (iteration index).
 - 2: Initial: $\bar{\mathbf{v}}^{(0)}$.
 - 3: **repeat**
 - 4: Calculate the conjugate gradient $\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}}^{(t)})$ by Theorem 3.
 - 5: Find step size $\mu^{(t)}$ by backtrack line search [34].
 - 6: Update $\bar{\mathbf{v}}^{(t+1)}$ as

$$\bar{\mathbf{v}}^{(t+1)} = \exp \left(j \arg \left(\bar{\mathbf{v}}^{(t)} + \mu^{(t)} \nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}}^{(t)}) \right) \right). \quad (34)$$
 - 7: Set $t = t + 1$ and calculate the secret key rate $\tilde{g}(\bar{\mathbf{v}}^{(t+1)})$.
 - 8: **until** $\frac{|\tilde{g}(\bar{\mathbf{v}}^{(t+1)}) - \tilde{g}(\bar{\mathbf{v}}^{(t)})|}{\tilde{g}(\bar{\mathbf{v}}^{(t)})} \leq \epsilon$.
-

C. Extraction of $\mathbf{P}_k$ From $\bar{\mathbf{P}}_k$

After obtaining the optimized $\bar{\mathbf{P}}_k$, which is denoted as $\bar{\mathbf{P}}_k^{\text{opt}}$, we need to extract the precoding matrix $\mathbf{P}_k$ from the variable $\bar{\mathbf{P}}_k^{\text{opt}}$. Specifically, the optimization problem is expressed as

$$\begin{aligned} & \underset{\mathbf{P}_k}{\text{minimize}} \quad \|\mathbf{I}_{M_U} \otimes \mathbf{P}_k - \bar{\mathbf{P}}_k^{\text{opt}}\|_F^2 \\ & \text{s.t.} \quad \text{C1.} \end{aligned} \quad (35)$$

Since the objective function in Problem (35) can be rewritten as $\sum_{l=1}^{M_U} \|\mathbf{P}_k - \bar{\mathbf{P}}_{k,l}^{\text{opt}}\|_F^2$, where $\bar{\mathbf{P}}_{k,l}^{\text{opt}} = [\bar{\mathbf{P}}_k^{\text{opt}}]_{(l-1)M_B:(l-1)M_B+(l-1)M_B:(l-1)M_B+(l-1)M_B}$, Problem (35) is a convex quadratic programming problem, which can be solved efficiently by using standard convex optimization program solvers, such as CVX [40].

The overall algorithm to solve the original problem (14) is summarized as Algorithm 3.

D. Complexity Analysis

We analyze the computational complexity of the proposed algorithm. We assume the number of the AO iteration is T_{AO} and then calculate the complexity required to solve each sub-problem. Specifically, for Algorithm 1, in each iteration, the complexity to update $\bar{\mathbf{P}}$ is $\mathcal{O}(KM^6)$. The complexity of evaluating the Lagrange multipliers $\lambda_i, i \in \{1, \dots, K\}$ can be ignored. Hence, the complexity for Algorithm 1 is $\mathcal{O}(T_{\text{KKT}}KM^6)$, where T_{KKT} is the required number of iterations. For Algorithm 2, the optimization of the RIS phase shifts depends on the number of gradient updates, T_{PGA} , and the amount of operations performed in each gradient update. The complexity for computing the gradient $\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}})$ is $\mathcal{O}((MNL)^3)$. Thus, the complexity of Algorithm 2 is

Algorithm 3 The Overall Algorithm for Problem (14)

- 1: Set: $\ell = 0$ (iteration index).
 - 2: **repeat**
 - 3: Update matrices $\bar{\mathbf{P}}_k^{(\ell)}, k \in \{1, \dots, K\}$ using Algorithm 1.
 - 4: Update the phase shifts $\bar{\mathbf{v}}^{(\ell)}$ using Algorithm 2.
 - 5: Set $\ell = \ell + 1$ and calculate the WSKR $\tilde{g}(\bar{\mathbf{v}}^{(\ell+1)})$.
 - 6: **until** $\frac{|\tilde{g}(\bar{\mathbf{v}}^{(\ell+1)}) - \tilde{g}(\bar{\mathbf{v}}^{(\ell)})|}{\tilde{g}(\bar{\mathbf{v}}^{(\ell)})} \leq \epsilon$.
 - 7: Recover the precoding matrices, $\mathbf{P}_k, k = 1, \dots, K$, from $\bar{\mathbf{P}}_k$ by solving Problem (35).
-

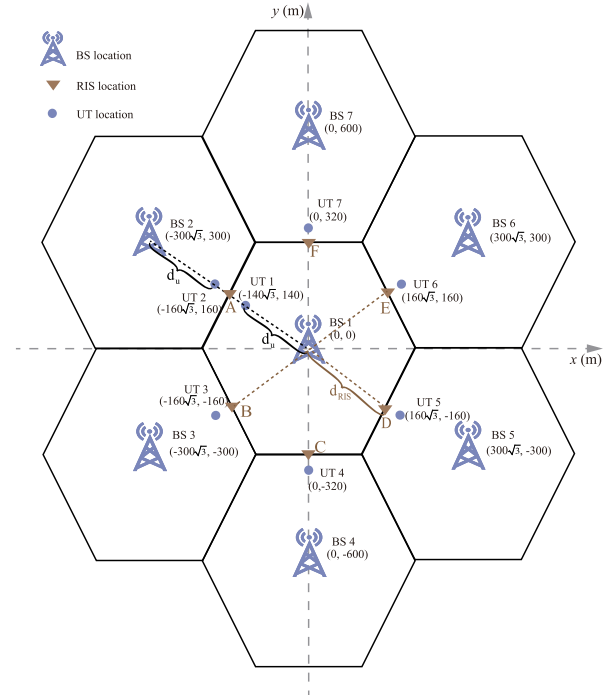


Fig. 3. Simulation setup for the RIS-aided PKG in a seven-cell scenario.

$\mathcal{O}(T_{\text{PGA}}(MNL)^3)$. Then, the complexity of the AO algorithm is $\mathcal{O}(T_{\text{AO}}(T_{\text{KKT}}KM^6 + T_{\text{PGA}}M^3N^3L^3))$. Additionally, the complexity to solve Problem (35) via interior-point methods is $\mathcal{O}(M \log \frac{M_B^2}{\epsilon})$ for an ϵ -accurate solution [34]. To sum up, the overall complexity of the proposed algorithm is $\mathcal{O}(T_{\text{AO}}(T_{\text{KKT}}KM^6 + T_{\text{PGA}}M^3N^3L^3) + M \log \frac{M_B^2}{\epsilon})$.

V. SIMULATION RESULTS

In this section, simulation results are presented to illustrate the performance of the proposed multi-cell RIS-aided PKG scheme.

As shown in Fig. 3, we consider a seven-cell scenario, where the x and y axes represent the horizontal plane. The coordinates of the BSs are $(0, 0)$, $(-300\sqrt{3}, 300)$, $(-300\sqrt{3}, -300)$, $(0, -600)$, $(300\sqrt{3}, -300)$, $(300\sqrt{3}, 300)$, and $(0, 600)$, respectively. The seven UTs are located at $(-140\sqrt{3}, 140)$, $(-160\sqrt{3}, 160)$, $(-160\sqrt{3}, -160)$, $(0, -320)$, $(160\sqrt{3}, -160)$, $(160\sqrt{3}, 160)$, and $(0, 320)$, respectively. Meanwhile, the RISs are located at the points A, B, C, D, E, and F, respectively, which are located at the six boundaries between the first cell and the remaining six

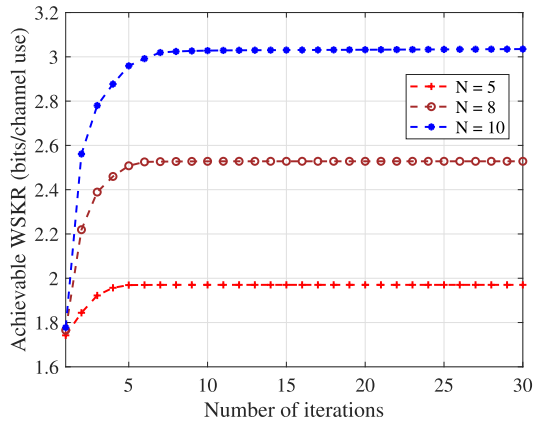


Fig. 4. The convergence of the AO algorithm for $M_B = 4$, $M_U = 1$, and $P_A = 30$ dBm.

cells, respectively. The heights of the BSs, RISs, and UTs are 30 m, 10 m, and 1.5 m, respectively.

In the simulation, the channels are modeled by the product of large-scale path loss and small-scale fading. In particular, the large-scale path loss is denoted as $\sqrt{\zeta_0}d^{-\alpha}$, where d , ζ_0 , and α are the distance, path loss at 1 m, and the path loss exponent, respectively. Due to the extensive obstacles and scatters between the BSs and the UTs, the path loss exponents of the BS-UT (BU) links, BS-RIS (BR) links, and the RIS-UT (RU) links are given by $\alpha_{BU} = 3.75$ and $\alpha_{BR} = \alpha_{RU} = \alpha_{RIS} = 2.2$, respectively [14]. Furthermore, the small-scale fading between the BSs and UTs is assumed to follow Rayleigh fading and the small-scale fading of the BS-RIS link and RIS-UT link are modeled as Rician distribution [33]. β denotes the Rician factor and is set as 3 in the simulation [14]. The noise power at the BSs and UTs is -90 dBm [33]. Also, the default weights, $w_1, \dots, w_k$, are set to all 1s. In the following results, the unit of WSKR has been converted to bits/channel use to align with the results in existing literature, e.g., [9], [15], [16], [17], [18], and [19].

Additionally, four baseline schemes are adopted: (1) **OP-Precoding, Rand-Phase**: the precoding matrices at the BSs are optimized by Algorithm 1 and the phase shifts at the RISs are randomly selected; (2) **OP-Precoding, No-RIS**: the precoding matrices at the BSs are optimized by Algorithm 1 and the coefficients of the RIS-related channels are set to zero; (3) **SC-Precoding, OP-RIS**: the precoding matrices at the BSs are the optimal precoding matrices in the single-cell scenario [37] and the phase shifts at the RISs are optimized by Algorithm 2; (4) **SC-Precoding, Rand-Phase**: the precoding matrices at the BSs are the optimal precoding matrices in the single-cell scenario [37] and the phase shifts at the RISs are randomly chosen.

First, Fig. 4 depicts the achieved WSKR versus the number of AO iterations. As can be observed, the WSKR converges rapidly to stationary values after a few iterations on average. For $N = 5$, the proposed algorithm converges after around 8 iterations on average. For the case with more RIS elements, i.e., $N = 10$, the number of iterations required for the convergence is 14 on average. This is because the solution space is enlarged when more optimization variables are involved and thus more iterations are required for convergence.

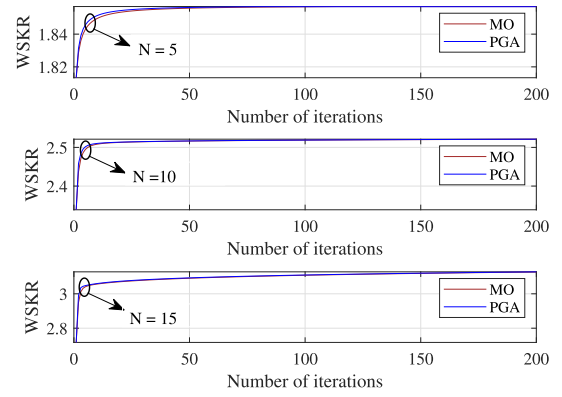


Fig. 5. Convergence behavior of the PGA and MO algorithms for $P_A = 30$ dBm, $M_B = 4$, and $M_U = 1$.

TABLE I
THE COMPUTATION TIME OF THE PGA ALGORITHM AND
THE MO ALGORITHM

	$N = 2$	$N = 4$	$N = 6$	$N = 8$	$N = 10$
MO	1.3125	28.0469	103.9844	502.9688	1017.6563
PGA	0.7812	25.5469	102.2031	500.3125	1012.8438

Then, for the optimization of phase shifts, we compare the performance of the adopted PGA algorithm with the existing manifold optimization (MO) algorithm [14], [41]. Specifically, we compare the performance of the acquired solutions and the complexity of the two algorithms. First, as shown in Fig. 5, the convergence values of the two algorithms are consistently identical for all the considered values of N , since they have a similar structure to achieve comparable performance levels. Then, the complexity of the PGA algorithm and the MO algorithm is determined by the complexity of each iteration and the number of iterations for convergence. In each iteration, the complexity of both algorithms is mainly dominated by the complexity of calculating the Euclidean gradient $\nabla_{\bar{\mathbf{v}}}^* g(\bar{\mathbf{v}}^{(t)})$, which is with a complexity order of $(MNL)^3$. However, in each iteration, the PGA algorithm distinguishes itself by offering simple computation steps, as it eliminates the need for projecting operations between Euclidean space and Riemannian space. Then, Fig. 5 shows that the convergence speed of the PGA algorithm and MO algorithm is nearly the same but the PGA algorithm converges a little faster compared to the MO algorithm. Thus, the overall complexity of the PGA algorithm is marginally lower than that of the MO algorithm. To further analyze the complexity, we provide the computation time of both algorithms for solving Problem (30). As shown in Table I, the running time of the PGA algorithm is slightly shorter than that of the MO algorithm. Overall, both algorithms can be applied to solve Problem (30).

Next, in Fig. 6, the WSKR versus the maximum transmit power at each BS for different schemes is plotted. First, it can be observed that the WSKRs in all of the schemes increase with the transmit power, since the impact of noise becomes less significant. At the same time, the WSKR increases with diminishing returns in the high transmit power region. This is because only the transmit power at the BSs is increased while the transmit power of the UTs is fixed that remains the system performance bottleneck. In particular, the channel estimations at the BSs are still impaired by the noise

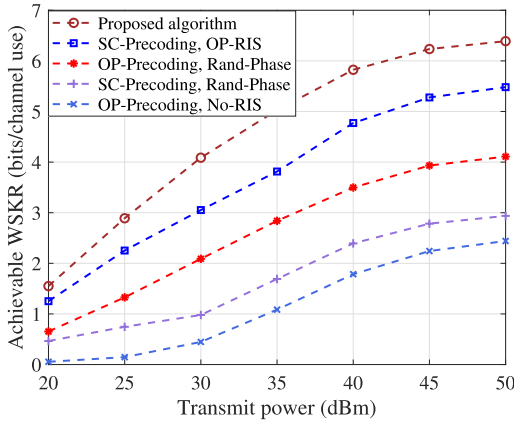


Fig. 6. Achievable WSKR versus the transmit power at the BSs, P_A , for $M_B = 4$, $M_U = 1$, and $N = 30$.

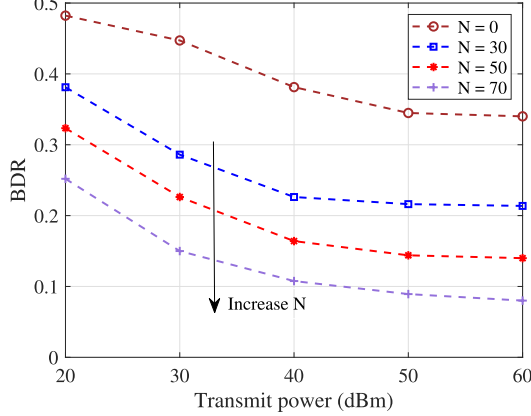


Fig. 7. BDR versus the transmit power P_A under the proposed algorithm for $M_B = 8$ and $M_U = 1$.

components, causing the saturation in the WSKR. Furthermore, it can be observed that the proposed algorithm outperforms the baseline schemes, since the precoding matrices of all the BSs and the phase shifts of the RISs are jointly optimized. By contrast, the SC-Precoding with OP-RIS scheme only optimizes the phase shifts but the precoding matrices of the BSs are not coordinately designed. In this case, the spatial diversity of the BSs is not fully exploited that results in the loss of WSKR. Then, the random phase shift schemes, i.e., the OP-Precoding with Rand-Phase and SC-Precoding with Rand-Phase schemes achieve lower WSKR, as the capability of controlling wireless environments of the RISs is not fully exploited. In addition, it can be seen that optimizing the RIS phase shifts achieves greater performance gains compared to optimizing the precoding matrices. This is because in Fig. 6, each RIS is equipped with 30 elements while each BS only has 4 antennas such that the former offers a higher flexibility to customize the end-to-end channels. Finally, it is noteworthy that the No-RIS scheme has the lowest WSKR and saturates at only 2.5 bits/channel use at the high transmit power regime. The reason behind this is that the non-reciprocal inter-cell interference dominates the estimation of both uplink and downlink channels that reduces the achievable WSKR. This observation emphasizes the necessity of deploying RIS in multi-cell PKG systems.

To illustrate the impact of the proposed algorithm on the channel reciprocity, we show the BDR result in Fig. 7.

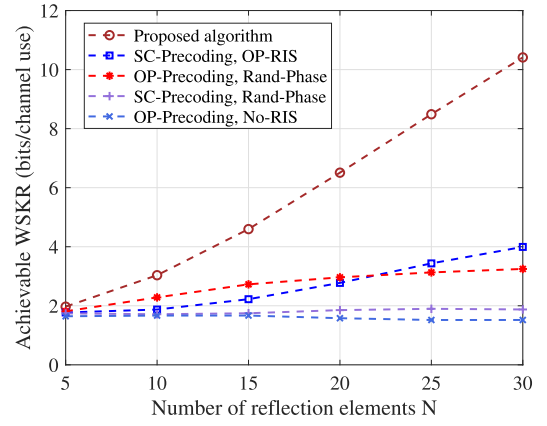


Fig. 8. Achievable WSKR versus the number of RIS elements N for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$.

Specifically, BDR is defined as the ratio between the number of disagreement bits and the number of total bits [2], [5], [6], [7]. It can be seen that the BDR of No-RIS, i.e., $N = 0$, is much higher than the BDR with RIS and the former saturates at around 0.34 in the high transmit power regime. This is because the non-reciprocal interference significantly degrades the similarity of the uplink and downlink channel estimations. By contrast, for $N > 0$, it can be seen that a lower BDR can be achieved with the increase of N , since the RIS is equipped with a higher ability to suppress the pilot contamination-caused inter-cell interference with larger N .

Fig. 8 shows the WSKR of different PKG schemes versus the number of elements at each RIS N . Firstly, as can be observed, the WSKR of the No-RIS scheme is significantly low because the UTs are located at the cell edge and suffer from severe inter-cell pilot contamination. In contrast, the WSKRs of all RIS-based schemes are improved with the number of RIS elements. Also, the proposed algorithm achieves the highest WSKR and the performance gain increases with N . This is because the multi-cell interference in the uplink and downlink is mitigated by adapting the phase shifts of the RIS and with more RIS elements in place, the proposed design enjoys a higher flexibility to create pencil-like beams to focus the reflected signals on the target UTs and BSs. In addition, it is noted that when $N \leq 20$, the WSKR of the SC-Precoding with OP-RIS scheme is less than that of the OP-Precoding with Rand-Phase scheme, while the SC-Precoding with OP-RIS scheme achieves a larger WSKR when $N \geq 25$. This is mainly because when N is small, the RIS's ability to alter the environment is relatively weak and thus, optimizing the phase shifts of the RIS contributes to less performance gain compared to optimizing the precoding matrices at the BSs. However, when N becomes larger, optimizing the RIS phase shifts can fully exploit its ability to mitigate the interference and achieve a higher WSKR.

Fig. 9 shows the WSKR versus the number of antennas at each BS or UT under the proposed algorithm. It can be seen that the WSKRs increase with the numbers of antennas equipped at the BS or UT, since more DoFs are introduced by higher dimensional channels for a larger M_B or M_U . In addition, as can be observed, increasing M_B can achieve a larger WSKR compared to increasing M_U . This is because

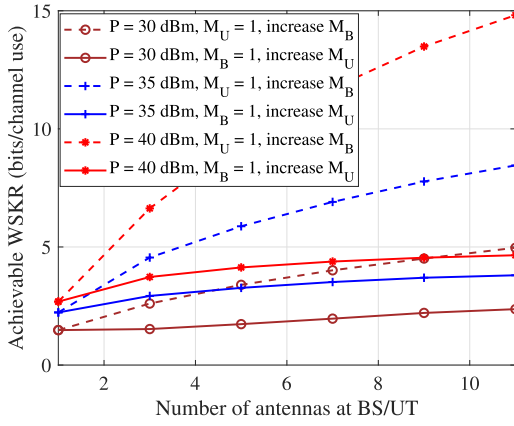


Fig. 9. Achievable WSKR versus the number of antennas at BS or UT for $N = 10$ and $P_A = 30$ dBm.

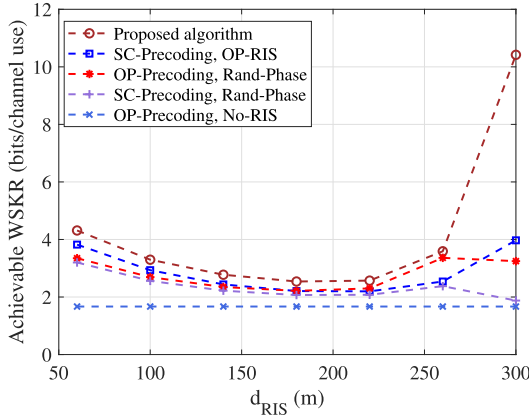


Fig. 10. Achievable WSKR versus the location of the RIS d_{RIS} for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$, and $N = 30$.

the precoding matrices at the BSs are jointly optimized and thus the increase of BS antennas can significantly enhance the beamforming capability to suppress the inter-cell interference.

Fig. 10 presents the WSKR versus the distance between the RISs and the center of the seven-cell cluster, i.e., d_{RIS} . Specifically, as shown in Fig. 3, each RIS is located along the line connecting the center point of the system to points A, B, C, D, E, and F, respectively, and we vary the location of the RISs from near the center point, $d_{\text{RIS}} = 60$ m, to the corresponding cell boundary, $d_{\text{RIS}} = 300$ m. Firstly, it can be observed that when $50 < d_{\text{RIS}} < 220$, the WSKRs of all the considered RIS-involved schemes decrease. This is mainly caused by the increased large-scale path loss of the RIS channels between BS 1 and UT 1. Specifically, the large-scale channel gain of the RIS channel is the product of the gain of the BS-RIS link and that of the RIS-UT link. Thus, for BS j and UT j , $j \in \{2, \dots, 7\}$, the large-scale fading of the RIS channels can be approximated by $\zeta_0 \sqrt{((600 - d_{\text{RIS}})(320 - d_{\text{RIS}}))^{-\alpha_{\text{RIS}}}}$, which is negligible for the considered $50 < d_{\text{RIS}} < 220$. However, for BS 1 and UT 1, the large-scale fading of the RIS channels can be approximated by $\zeta_0 \sqrt{(d_{\text{RIS}}(280 - d_{\text{RIS}}))^{-\alpha_{\text{RIS}}}}$, which decreases significantly when the RIS moves away from BS 1. Additionally, when $220 \leq d_{\text{RIS}} \leq 300$, it can be observed that the Rand-Phase baselines achieve their maximum value when the RIS is located near the UTs, i.e., $d_{\text{RIS}} = 260$ m,

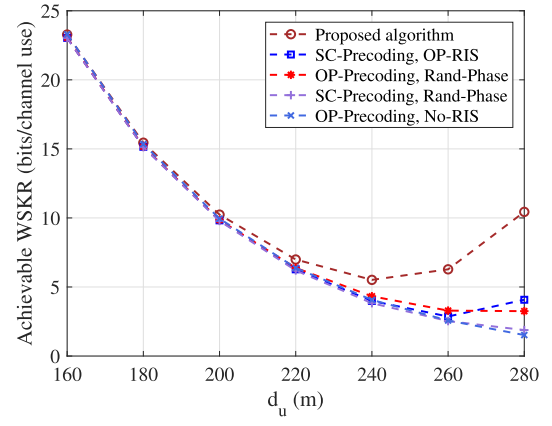


Fig. 11. Achievable WSKR versus the location of the UT for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$, and $N = 30$.

which is caused by the array gain brought by the RIS. On the other hand, for the OP-RIS schemes, the maximal WSKR is achieved when $d_{\text{RIS}} = 300$ m, since the multi-cell interference can be significantly reduced by the optimized RIS deployed at the cell boundaries. Finally, the proposed algorithm enjoys superior performance than that of the four baseline schemes in all of the considered cases and the maximum WKR can be achieved when $d_{\text{RIS}} = 300$ m, because deploying the RIS close to the cell boundaries is more effective in alleviating the negative impacts caused by the inter-cell pilot contamination.

Fig. 11 shows the WSKR versus the location of the UTs under different schemes, where d_u denotes the distance between the BS and UT in each cell. First, it can be observed that the WSKRs of the Rand-Phase schemes and No-RIS schemes decrease with d_u . This is mainly due to the following two reasons. First, the channel gain between the BSs and UTs decreases as the UTs move away from their home BSs. Second, the inter-cell interference becomes dominated when the UTs move from the cell center to the cell boundary, while the Rand-Phase or No-RIS schemes cannot effectively suppress this interference. In contrast, the optimized RIS schemes, i.e., the proposed algorithm and the SC-Precoding with OP-RIS scheme, achieve larger WSKRs when the UTs are located close to the cell boundary, i.e., $d_u = 280$ m, because the inter-cell interference is substantially mitigated by the RISs. Finally, the proposed algorithm outperforms the baseline schemes for all considered values of d_u and provides the significant performance gain when the RISs are close to the cell boundary.

Furthermore, since the amplitude of the reflected signals by each RIS element may be attenuated due to the energy absorption of phase shifters in practical systems, we study the impact of the amplitude variations of the reflection on WSKR performance. Specifically, we represent the phase-shifting vector $\mathbf{v}$ as $\eta[e^{j\theta_1}, \dots, e^{j\theta_{NL}}]^T$, where θ_n and η denote the phase of the n -th RIS element and the reflection amplitude. Then, it can be seen from Fig. 12 that the WSKR increases with the reflection amplitude since the signal power is enhanced under a larger η . Note that when $\eta = 0.2$, compared to the No-RIS scheme, the proposed algorithm can only achieve 0.6 bits/channel use performance gain. By contrast, when the reflection amplitude is 1, 10 bits/channel use performance gain

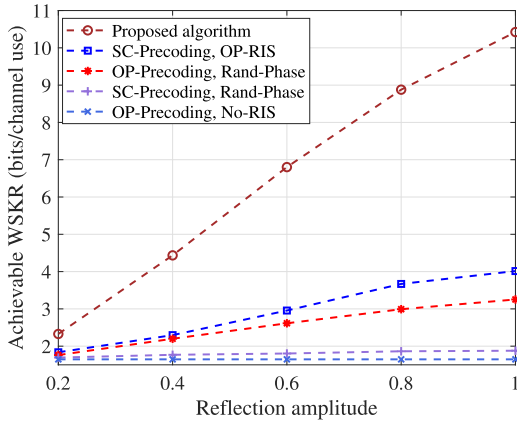


Fig. 12. Achievable WSKR versus the reflection amplitude, η , for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$, and $N = 30$.

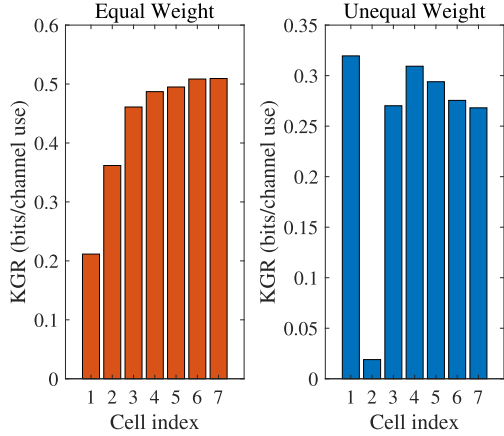


Fig. 13. Individual KGR under two sets of weights for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$, and $N = 10$.

can be achieved. This shows that the reflection amplitude of the RIS elements has a significant impact on the WSKR performance.

Next, we show the impact of the weights on KGR performance. The weight w_k represents the priority of the k -th cell. In particular, the KGR between BS k and UT k can be increased by increasing the ratio $w_k / \sum_{i=1}^K w_i$. To illustrate this point, as shown on the left-hand side subfigure of Fig. 13, when equal weights are adopted, the KGR of the first cell is much lower than that of other cells since BS 1 is located at the center of the considered seven-cell system and heavily interfered by the UTs from other cells. Then, we investigate the impacts of unequal weights by considering: $w_1 = 0.909$, $w_2 = \dots = w_7 = 0.0152$, where w_1 is much higher than $w_j, \forall j \neq 1$. In this setting, as can be seen from the right-hand side subfigure of Fig. 13, the KGR of the first cell is greatly increased under unequal weights. Meanwhile, it can be seen that the second cell has the lowest KGR because UT 1 is close to UT 2 that introduces more interference to the second cell. Meanwhile, although RIS 1 is located at the boundary of the first cell and the second cell, the weight of the first cell is much larger than that of the second cell, which causes marginal performance improvement for the second cell.

Then, we compare the PKG performance by deploying a single RIS and multiple RISs. For the single RIS case, we assume that the RIS is located at point A to simultaneously

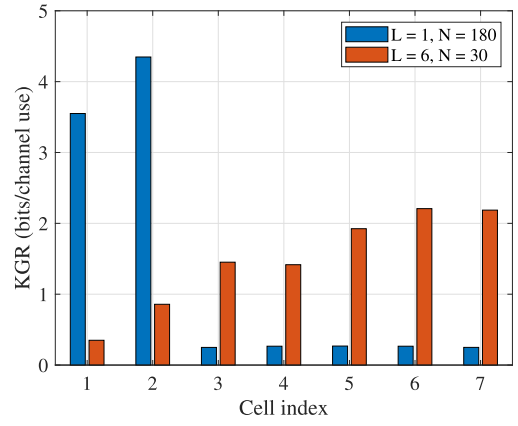


Fig. 14. Individual KGR under different RIS deployment for $P_A = 30$ dBm, $M_B = 4$, $M_U = 1$.

TABLE II
NIST RANDOM TEST RESULT

	Pass ratio	P-value
Approximate entropy	0.9911	0.4851
Runs	0.9985	0.5088
Ranking	0.9941	0.4992
Longest runs of ones	0.9899	0.6238
Frequency	1	0.5201
FFT	0.9853	0.4295
Block frequency	1	0.4401
Cumulative sums	0.9981	0.5057
Serial	0.9838	0.5013, 0.4895

facilitate the first cell and the second cell with $N = 180$. For the multiple RISs case, the six RISs are located at points A, B, C, D, E, F, respectively, and each RIS is equipped with $N = 30$ elements. The simulation results are shown in Fig. 14. First, as can be observed, in the single RIS case, the KGR of the first cell and second cell is greatly improved while the KGRs of other cells are much lower. This means that deploying a single RIS with a large number of elements can effectively improve the KGR performance of its nearest cells, while the BSs and UTs in other cells suffer from severe power loss due to the long communication distance between them and the RIS. In contrast, it can be seen that in the multiple RISs case, the KGRs of most cells are improved. This is because when multiple RISs are deployed in the network, the distance between each UT and its nearest RIS is reduced due to spatial diversity, which thus increases the channel gain. Meanwhile, the minimum KGR increases from 0.25 bits/channel use of the single RIS case to 0.35 bits/channel use. Furthermore, the WSKRs of the single RIS case and multiple RISs case are 9.19 bits/channel use and 10.39 bits/channel use, respectively. These results validate the beneficial effects of multiple RISs on the improvement of both the WSKR and the KGR-fairness between cells.

Finally, to verify the randomness of the obtained bit sequences for cryptographic applications, we conduct the National Institute of Standards and Technology (NIST) randomness test [42] on the quantized bits. Note that NIST is widely adopted to evaluate the randomness of true-random and pseudo-random number generators. The output of the NIST is p-values, which is examined to ensure uniformity. The tested sequence is considered to pass the test if the p-value is greater

than 0.01. In the simulation, we perform 9 kinds of NIST items for 10,000 trials, with the length of each sequence being 2048 bits. The test results are shown in Table II, where the pass ratio represents the number of passed trials over the 10,000 trials. As can be observed, the p-values of all the test items are significantly greater than 0.01, which means the sequence can be considered to be uniformly distributed. Also, the pass ratios are all higher than 98%. This confirms the excellent randomness of the bits generated by the proposed PKG scheme.

VI. CONCLUSION

In this paper, we incorporated RISs in multi-cell PKG systems to alleviate the negative impact of multi-cell pilot contamination on PKG performance. Specifically, we studied the WSKR maximization problem by jointly optimizing the precoding matrices at the BSs and the phase shifts at the RISs. To tackle this non-convex problem, we denoted the Kronecker product of the identity matrix and the precoding matrix as the equivalent matrix, and then applied an AO-based algorithm that alternatively solves the sub-problem for the equivalent matrices and the sub-problem for phase-shifting vector. In particular, a Lagrangian dual algorithm and a PGA algorithm were employed to design the equivalent matrices and the phase-shifting vector, respectively. Simulation results verified that the proposed algorithm outperforms the baseline schemes. Moreover, the higher WSKR and lower BDR can be observed with the increase of RIS elements number. Finally, a single RIS can only improve the PKG performance of its nearest cells, while distributed RISs can effectively facilitate more cells.

APPENDIX

A. Proof of Theorem 1

The KGR between BS k and UT k is expressed as [43] and [44]

$$R_k = \mathcal{I}(\mathbf{y}_k; \mathbf{z}_k) = \ln \frac{\det(\mathbf{R}_{\mathbf{y}_k} \mathbf{R}_{\mathbf{z}_k})}{\det(\mathcal{R}_{\mathbf{y}_k \mathbf{z}_k})}, \quad (36)$$

where the covariance matrices are denoted as $\mathbf{R}_{\mathbf{y}_k} = \mathbb{E}\{\mathbf{y}_k \mathbf{y}_k^H\}$, $\mathbf{R}_{\mathbf{z}_k} = \mathbb{E}\{\mathbf{z}_k \mathbf{z}_k^H\}$, and

$$\mathcal{R}_{\mathbf{y}_k \mathbf{z}_k} = \begin{bmatrix} \mathbf{R}_{\mathbf{y}_k} & \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \\ \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} & \mathbf{R}_{\mathbf{z}_k} \end{bmatrix}. \quad (37)$$

Employing the determinant of a block matrix, we have

$$\det(\mathcal{R}_{\mathbf{y}_k \mathbf{z}_k}) = \det(\mathbf{R}_{\mathbf{y}_k}) \det(\mathbf{R}_{\mathbf{z}_k} - \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} \mathbf{R}_{\mathbf{y}_k}^{-1} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k}). \quad (38)$$

Thus, the KGR is reformulated as

$$R_k = \ln \left(\frac{\det(\mathbf{R}_{\mathbf{y}_k}) \det(\mathbf{R}_{\mathbf{z}_k})}{\det(\mathbf{R}_{\mathbf{y}_k}) \det(\mathbf{R}_{\mathbf{z}_k} - \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} \mathbf{R}_{\mathbf{y}_k}^{-1} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k})} \right) \quad (39)$$

$$= -\ln \det(\mathbf{I} - \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k} \mathbf{R}_{\mathbf{y}_k}^{-1} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} \mathbf{R}_{\mathbf{z}_k}^{-1}). \quad (40)$$

Given the channel estimations in (7) and (8), the channel covariance matrices are calculated as

$$\mathbf{R}_{\mathbf{z}_k} = \bar{\mathbf{P}}_k \left[\sum_{j=1}^K \mathbf{R}_{k,j}^d / M_U + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \right]$$

$$\times \sum_{j=1}^K \mathbf{R}_{k,j}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M) / M_U + \mathbf{I}_M \Big] \bar{\mathbf{P}}_k^H, \quad (41)$$

$$\begin{aligned} \mathbf{R}_{\mathbf{y}_k} &= \sum_{i=1}^K \bar{\mathbf{P}}_i (\mathbf{R}_{i,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \\ &\times \mathbf{R}_{i,k}^r (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M)^H) \bar{\mathbf{P}}_i^H + \mathbf{I}_M, \end{aligned} \quad (42)$$

$$\begin{aligned} \mathbf{R}_{\mathbf{y}_k \mathbf{z}_k} &= \bar{\mathbf{P}}_k (\mathbf{R}_{k,k}^d + (\bar{\mathbf{v}}^H \otimes \mathbf{I}_M) \\ &\times \mathbf{R}_{k,k}^r (\bar{\mathbf{v}} \otimes \mathbf{I}_M)) \bar{\mathbf{P}}_k^H / \sqrt{M_U} = \mathbf{R}_{\mathbf{z}_k \mathbf{y}_k}. \end{aligned} \quad (43)$$

Substituting these channel covariance matrices into (40) and the result follows immediately.

B. Proof of Proposition 1

First, the eigenvalue decomposition of $\mathbf{R}_{k,j}^d$ can be described as $\mathbf{Q}_D \text{diag}(\mathbf{x}) \mathbf{Q}_D^H$, where $\mathbf{x} = [x_1, \dots, x_M]$ is the eigenvalue vector and $\mathbf{Q}_D = [\mathbf{c}_1, \dots, \mathbf{c}_M]$ is the eigenmatrix and $\mathbf{c}_\ell, \ell \in \{1, \dots, M\}$, is the eigenvector. By denoting $\sum_{j=1}^K P_j \mathbf{R}_{k,j}^d / M_B + M_U \mathbf{I}_M = \mathbf{A}$ and $\sum_{i=1}^K P_i \mathbf{R}_{i,k}^d / M_B + \mathbf{I}_M = \mathbf{B}$, the derivative of R_k w.r.t. $x_\ell, \ell \in \{1, \dots, M\}$, is

$$\begin{aligned} \partial R_k / \partial x_\ell &= -\text{tr} \left(\mathbf{B}^{-1} \mathbf{R}_{k,k}^d (\mathbf{A} - P_k / M_B \mathbf{R}_{k,k}^d \mathbf{B}^{-1} \mathbf{R}_{k,k}^d)^{-1} \right. \\ &\times P_k / M_B \mathbf{R}_{k,k}^d \mathbf{B}^{-1} P_i / M_B \mathbf{c}_\ell \mathbf{c}_\ell^H \Big). \end{aligned} \quad (44)$$

Since $\mathbf{A} - P_k / M_B \mathbf{R}_{k,k}^d \mathbf{B}^{-1} \mathbf{R}_{k,k}^d \succ \mathbf{0}$ and $\mathbf{c}_\ell \mathbf{c}_\ell^H \succeq \mathbf{0}$, we have $\partial R_k / \partial x_\ell < 0$. Thus, R_k decreases monotonically with x_ℓ . When $\lambda_\ell(\mathbf{R}_{k,j}^d) \rightarrow \infty$, we have $\lambda_\ell(\sum_{i=1}^K P_i / M_B \mathbf{R}_{k,j}^d + \mathbf{I}_M) \rightarrow 0$, and thus $R_k \rightarrow 0$. The proof for $\mathbf{R}_{i,k}^d$ and P_i can be derived similarly. This completes the poof.

C. Proof of Theorem 3

We can calculate the conjugate gradient of $\mathcal{L}(\bar{\mathbf{P}}, \Lambda)$ w.r.t. $\bar{\mathbf{P}}$ as

$$\begin{aligned} \frac{\partial \mathcal{L}(\bar{\mathbf{P}}, \Lambda)}{\partial \bar{\mathbf{P}}^*} &= \sum_{k=1}^K w_k \left((\bar{\mathbf{P}} \mathbf{M}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)^{-1} \bar{\mathbf{P}} \mathbf{M}_k \right. \\ &\left. - (\bar{\mathbf{P}} \mathbf{N}_k \bar{\mathbf{P}}^H + \mathbf{I}_M)^{-1} \bar{\mathbf{P}} \mathbf{N}_k \right) - \bar{\mathbf{P}} \Lambda. \end{aligned} \quad (45)$$

Set $\frac{\partial \mathcal{L}(\bar{\mathbf{P}}, \Lambda)}{\partial \bar{\mathbf{P}}^*} = \mathbf{0}$ and take the $\text{vec}(\cdot)$ operation on both sides of the equation, we can obtain the KKT condition as shown in (17). This completes the proof.

D. Proof of Theorem 4

First, the KGR is given by

$$R_k = \ln \det(\mathbf{R}_{\mathbf{y}_k}) - \ln \det(\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k}) \quad (46)$$

and the differential of R_k w.r.t. $\bar{\mathbf{v}}$ is written as

$$\begin{aligned} d(R_k) &= \text{tr}(\mathbf{R}_{\mathbf{y}_k}^{-1} d(\mathbf{R}_{\mathbf{y}_k})) - \text{tr}((\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \\ &\times \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} d(\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})). \end{aligned} \quad (47)$$

Then, the first term of (47) is calculated as

$$\text{tr}(\mathbf{R}_{\mathbf{y}_k}^{-1} d(\mathbf{R}_{\mathbf{y}_k}))$$

$$= \text{tr} \left(\mathbf{K}_{MNL} \sum_{i=1}^K (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i) \mathbf{R}_{i,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i)^H \right. \\ \left. \times (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \mathbf{R}_{\mathbf{y}_k}^{-1} \mathbf{K}_M (\mathbf{I}_M \otimes d(\bar{\mathbf{v}}^H)) \right). \quad (48)$$

Similarly, we can calculate the second term of (47) as

$$\text{tr} \left((\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} \right. \\ \left. \times d(\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k}) \right) \quad (49) \\ = \text{tr} \left(\mathbf{K}_{MNL} \sum_{i=1}^K (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i) \mathbf{R}_{i,k}^r (\mathbf{I}_{NL} \otimes \bar{\mathbf{P}}_i)^H (\bar{\mathbf{v}} \otimes \mathbf{I}_M) \right. \\ \left. \times (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} \mathbf{K}_M (\mathbf{I}_M \otimes d(\bar{\mathbf{v}}^H)) \right) \\ - \text{tr} \left(\mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k} (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} d(\mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k}) \right) \\ - \text{tr} \left(\mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k} (\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} d(\mathbf{R}_{\bar{\mathbf{z}}_k}^{-1}) \right) \\ - \text{tr} \left((\mathbf{R}_{\mathbf{y}_k} - \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} \mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k})^{-1} \mathbf{R}_{\mathbf{y}_k \bar{\mathbf{z}}_k} \mathbf{R}_{\bar{\mathbf{z}}_k}^{-1} d(\mathbf{R}_{\bar{\mathbf{z}}_k \mathbf{y}_k}) \right). \quad (50)$$

By calculating each differential term in (50), we can obtain $d(R_k) = \text{tr}(\mathbf{G}_k d(\mathbf{I}_M \otimes d(\bar{\mathbf{v}}^H)))$. Therefore, the conjugate gradient of R_k w.r.t. $\bar{\mathbf{v}}$ is calculated as $\frac{\partial R_k}{\partial \bar{\mathbf{v}}^*} = \sum_{m=1}^M \bar{\mathbf{g}}_{k,m}$. This completes the proof.

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